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YIELDVISION: MACHINE LEARNING FOR CROP YIELD PREDICTION

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Abstract

YieldVision is a machine-learning decision-support platform that predicts **sugarcane yield** under climatic stress. Four regressors (Linear Regression, Decision Tree, Random Forest and SVR) are trained on **120 multi-varietal field trials** with 11 agronomic, weather and management features and combined into a soft-weighted **Voting Regressor**. The NOAA Oceanic Niño Index (ONI) grades El Niño crop-risk, and a FastAPI + React dashboard in 12 languages delivers predictions, What-If scenarios and advisories.

Background

- Sugarcane sustains about 5 crore farmers and dependents in India and feeds the ethanol programme
- Its 10–14-month cycle exposes it to El Niño rainfall deficits and heat stress
- Survey and single-model forecasts ignore ENSO teleconnections and fail in anomalous years
- Many research prototypes never reach farmers for lack of an accessible, multilingual interface
- Early forecasts help farmers, sugar mills, insurers and policy makers plan ahead

Objectives

- Predict sugarcane yield reliably under varying climatic conditions
- Assess the influence of El Niño / ENSO stress using the ONI
- Benchmark LR, Decision Tree, Random Forest and SVR regressors
- Test whether a Voting Regressor is more robust than any single model
- Deploy a multilingual dashboard with What-If simulation, advisories and PDF reports

Methodology

Curate the 120-trial benchmark (11 features; target in quintal/acre) and preprocess it with a ColumnTransformer (standard scaling + one-hot encoding); split 80:20 (96 train / 24 test); train LR, DT, RF (100 trees) and SVR (RBF, C = 100, ε = 1.0); combine them as $\hat{y} = 0.45 \cdot SVR + 0.25 \cdot DT + 0.20 \cdot RF + 0.10 \cdot LR$; validate with hold-out, 5-fold CV, duplicate-aware folds and a weight-sensitivity study; classify the ENSO phase and risk level from the ONI (El Niño ≥ +0.5 °C, La Niña ≤ −0.5 °C); and serve results through FastAPI with SQLite persistence, Gemini advisories and a React dashboard.

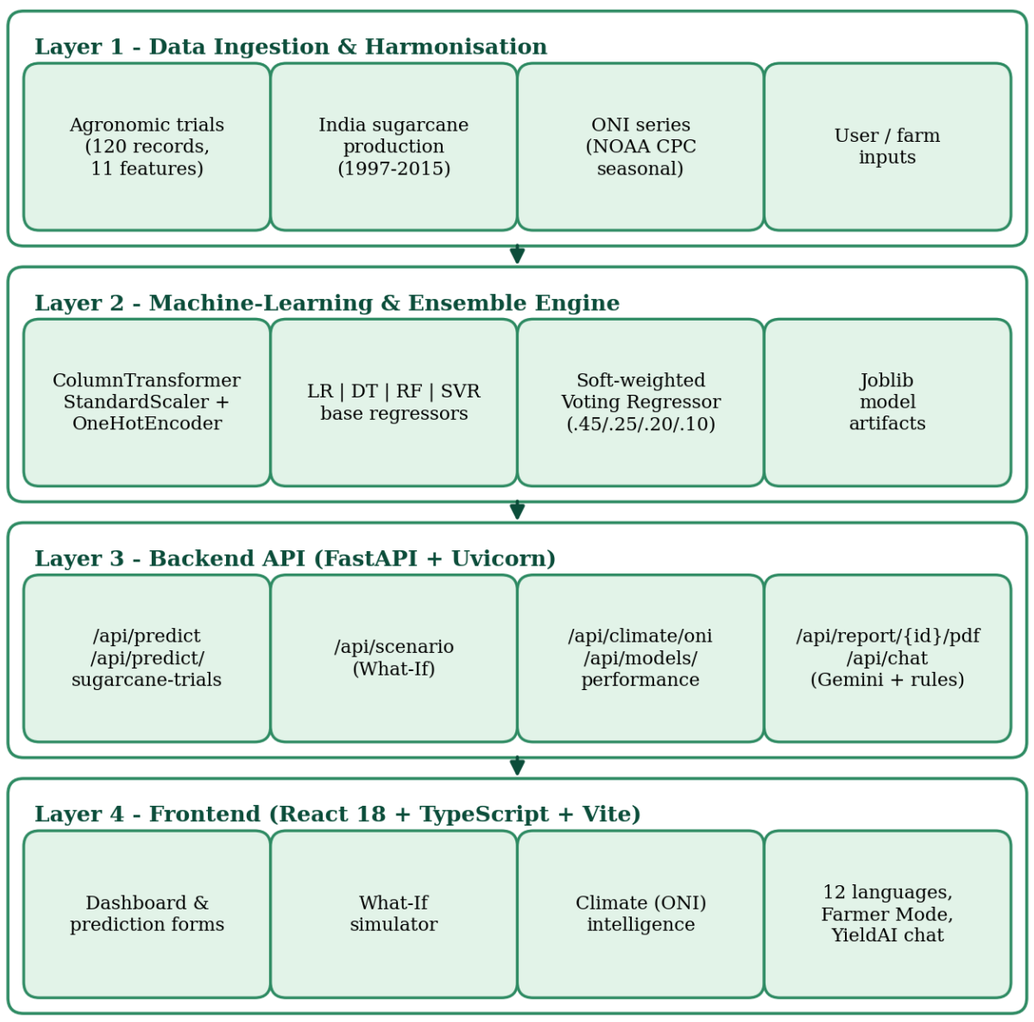


Figure 1: System Design (four-layer architecture)

Results and Discussion

Accuracy: On the 24-record hold-out set the Voting Regressor reaches R^2 = 0.9993, RMSE = 2.79 quintal/acre (≈ 0.69 t/ha, 0.7% of mean yield) and MAE = 2.12; SVR is marginally better on this single split (R^2 = 0.9997).
Stability: Under 5-fold cross-validation the ensemble has the lowest RMSE (3.34) and the smallest fold-to-fold R^2 spread (SD 0.0003); its weights cut RMSE by about 36% versus equal weighting.
Drivers: Nitrogen, crop duration, phosphorus, potassium and pest control account for ~81% of Random-Forest importance; omitting pest control lowers predicted yield by 27.6%.
Climate risk: ONI grading escalates risk from Moderate to High when El Niño (+1.8 °C) coincides with only 2 irrigations/month.
Limitations: Scores reflect in-distribution trial plots (120 records, 25% duplicates); weather What-If responses are indicative only.

Model	Hold-out RMSE	Hold-out R^2	5-fold CV RMSE	5-fold CV R^2
Linear Regression	16.82	0.9754	13.22	0.9866
Decision Tree	2.47	0.9995	3.74	0.9989
Random Forest	4.26	0.9984	6.57	0.9967
SVR (RBF)	1.98	0.9997	3.53	0.9990
Voting Regressor	**2.79**	**0.9993**	**3.34**	**0.9991**

Table 1: Hold-out (n = 24) and 5-fold CV performance, quintal/acre

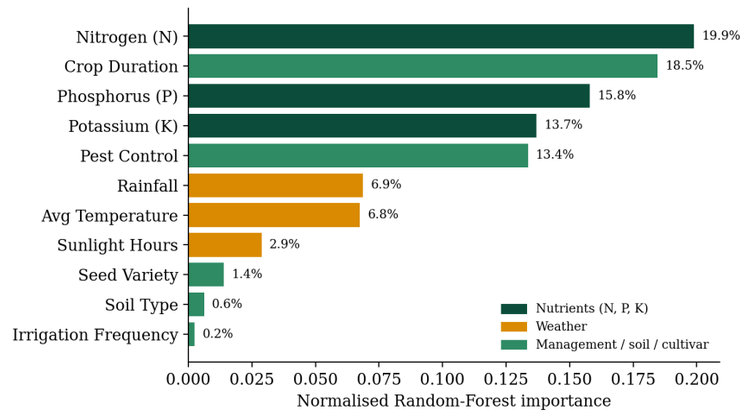


Figure 2: Random-Forest feature importance

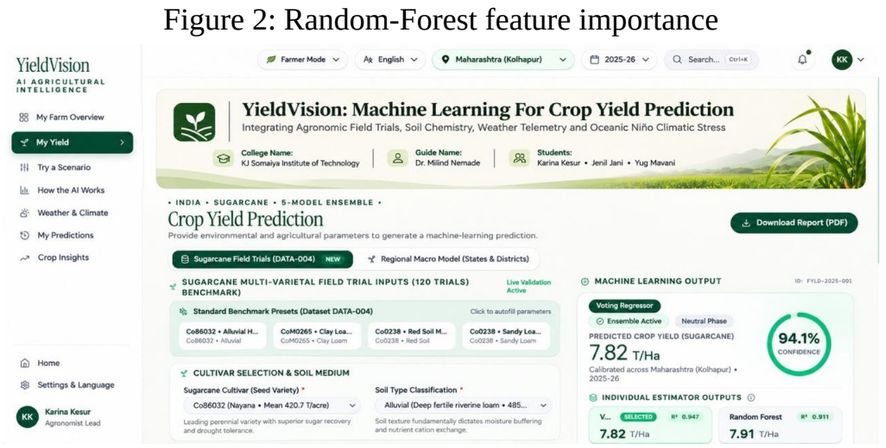


Figure 3: YieldVision dashboard (Farmer Mode, Kolhapur, Maharashtra)

Conclusion

YieldVision meets its objectives by combining agronomic trial data, weather inputs and ONI climate information in a soft-weighted ensemble that is the most dependable estimator rather than the winner on every split (CV RMSE 3.34 quintal/acre; $R^2 \geq 0.998$ even with duplicates removed). Delivered as a 12-language web platform, it supports farmers, sugar mills, insurers and policy makers. Future work: learn ONI/year as model inputs, add prediction intervals, and integrate IoT sensor and satellite (NDVI) data.

Acknowledgement

This research work is supported by K. J. Somaiya Institute of Technology.

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